V2AI: AI Lecture Video Understanding System

M25CSA010: Jyoti Dwivedi

M25CSA018: Mamta Chauhan

Abstract

- This project aims to develop an AI system that can understand and analyze video lectures, generating summaries and key insights to enhance learning experiences.

Materials

- Dataset: TED Talks
- Libraries: FastAPI, PyTorchs, Pydantic, Huggingface Transformes
- Frontend: StreamLit
- Containerization & Deployement: Docker, Huggingface
- Deployment: Google Cloud

Introduction

• Automated Lecture Processing: Converts uploaded lecture videos into structured, actionable educational content through speech-to-text and AI-powered analysis pipelines.

• Multi-Format Output Generation: Automatically creates transcripts, summaries, key concept extractions, flashcards, and quiz questions from a single video source.

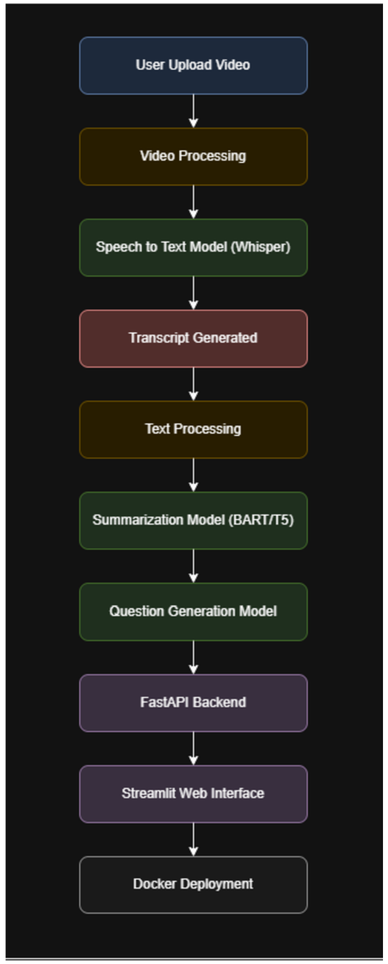
• Advanced NLP Pipeline: Utilizes Whisper for accurate transcription, Transformers for summarization, and domain-specific models for concept extraction from technical content.

• End-to-End Deployment Architecture: Implements complete operational infrastructure with FastAPI backend, Streamlit frontend, Docker containerization, and scalable cloud deployment on HuggingFace Spaces.

• Production-Ready Operations: Focuses on real-world MLOps challenges including data ingestion pipelines, error handling, storage optimization, user traffic management, and server-side monitoring.

• Demonstrable Educational Impact: Delivers 90-95% accuracy system capable of reducing student note-taking time while providing searchable, interactive study materials accessible from any device.

Workflow Diagram



Methodology

Data Ingestion & Processing Pipeline:

- Video upload via Streamlit frontend and FastAPI backend.
- OpenAI Whisper converts audio to transcript

NLP-based Knowledge Extraction:

- HuggingFace T5/BART for lecture summarization.
- Sentence-BERT for keyword and concept extraction.
- LangChain manages the processing pipeline and context flow.

Containerization & Deployment

- All components containerized using Docker.
- Services include Whisper, NLP module, FastAPI backend, Streamlit UI.

Database & Storage Strategy

- PostgreSQL schema for transcripts, summaries, quizzes, and user sessions.
- Maintain file versioning and audit logs.
- Implement user access control and automated data cleanup.